The Black Hole Information Paradox - A Void of Consensus

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1 Introduction

Black holes sit at the intersection of general relativity (GR) and quantum mechanics (QM), two theories that have been well-tested and proved on their own, yet cannot seem to get along together. But what is a black hole? A black hole is created when a supermassive star collapses, gravity beating the outward radiation pressure and a large amount of mass falls into a very small volume, causing a black hole to form. Black holes also have an event horizon, and following the GR equivalence principle, if an observer were to fall in, they would notice nothing strange around them when passing through the horizon, just like one would notice nothing strange in free fall in any other gravitational well[1].

This classical description works reasonably well for most calculations, however when we try to marry it with QM, many things start to go wrong and this is where the paradox comes to light. In Stephen Hawking's calculation of the temperature of the black hole, he noticed it was only reliant on the macroscopic properties like the mass, charge, and angular momentum. Additionally, he followed that up with showing that black holes produce radiation, specifically thermal in nature[2].

This becomes a problem when infalling objects have pure quantum states, while thermal radiation is a mixed state and when the black hole evaporates completely, the information is lost forever[3]. This violates unitarity, a basic and fundamental principle in QM. This contradiction is known as the black hole information paradox and suggests that at least one of our core methods, QM, GR, or the semiclassical approximation that Hawking used in his calculation, must break down.

2 Classical and Quantum Foundations

2.1 Black Holes in General Relativity

The classical solution begins with Einstein's field equations for a black hole with mass M and zero angular momentum. The Schwarzschild's spacetime metric is written as:

$$ds^2 = -\left(1 - \frac{2MG}{rc^2}\right)c^2dt^2 + \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad \text{with} \tag{1}$$

$$d\Omega^2 = d\theta^2 + sin^2\theta d\phi^2 \tag{2}$$

There is a beautiful derivation in reference [4]. We can set the speed of light c to be unity, and G is the gravitational constant. The event horizon occurs when $r = R_s = 2GM$, or the Schwarzschild radius, and separates the universe into two regions. The exterior of the black hole, where light can escape into infinity, and the interior, where light is lost forever. There are two singularities when looking at equation 1, the first at $r = 0$, and the second at $r = 2GM$. The latter one can be fixed by using a change of coordinates, but the first one is a problem, the black hole singularity. Additionally, the second term flips signs at the Schwarzschild radius, indicating the space-like and time-like coordinates flip. If one were in

the black hole, the singularity wouldn't be a position in space, but more so something in your future[1], [4].

The Schwarzschild metric is true for any spherically symmetric vacuum solution to Einstein's field equations. As $M \to 0$ we recover Minkowski space, as expected. We also get increasingly Minkowskian as $r \to \infty$, and this property is known as asymptotic flatness, where distant objects can be treated as unaffected by the gravity of the massive object, an isolated system [4]. This shows that all vacuum spherically symmetric spacetimes are static, and can be uniquely described by the Schwarzschild metric, this being known as Birkhoff's theorem. This is important as it is one of the first uniqueness theorems in GR, and can be used to describe the gravitational field of a spherically symmetric body like a star, even if it pulses or changes size, the field stays identical to that of a static, spherical mass. It also implies that the vacuum around a spherical object is static[4].

Outside the horizon, the Schwarzschild metric describes a non-rotating black hole, or a stationary black hole. In the case of a rotating spherical object, the Kerr-Newman metric is used for a stationary black hole[4]. Both of these types can be described using the no-hair theorem, which states that a black hole is characterized by only it's mass M, charge Q, and angular momentum J. "No-hair" comes from there being no additional features like magnetic moments of specific structures, or "hair", that can be used to differentiate one black hole from another. This description compares black holes to be similar to fundamental particles, regardless of how complex the matter they consume is[4], [5].

2.2 Quantum Mechanics and Information

In the QM description, we can start with understanding a pure versus mixed state. A pure quantum state is one where we have the most knowledge about the state of a quantum system. It can be described by a single wavefunction,

$$|\psi\rangle = \sum_i c_i |\phi_i\rangle \tag{3}$$

and all the uncertainty is the intrinsic quantum uncertainty due to the Heisenberg principle. In this state, the entropy is zero and the trace of the density matrix squared is equal to one[3].

$$Tr(\rho^2) = 1 \quad \text{(Pure State)} \tag{4}$$

The density matrix is used to calculate the probabilities of the outcomes of measurements in physical systems, with probabilities as the diagonal elements, and the phase information in the off-diagonal elements[3].

$$\rho = |\psi\rangle \langle\psi| \tag{5}$$

A mixed state on the other hand, represents a classical uncertainty about what state the quantum system is in. This is different from a superposition, which is considered a pure state that can be written in terms of other pure states with the phase relationships between

the components known, such as in equation 3. A mixed state is a statistical ensemble of the different pure states without the phase coherence, and the exact state can be unknown due to incomplete information, noise, or environmental factors. It cannot be defined through the typical Hilbert space vector $|\psi\rangle$, but only the density matrix ρ, which includes probabilities that represent the classical uncertainty in the system. The trace of the density matrix squared is also less that one, unlike in a pure state[3].

$$Tr(\rho^2) < 1 \quad \text{(Mixed State)} \tag{6}$$

Unitarity also plays an important role in both QM and in the paradox, but what is it? Quantum evolution is governed by a unitary operator U, which when applied to a wavefunction, applies a symmetric operation like a rotation or shift in Hilbert space, but preserves the norm, any information, and the entropy in a closed system. This means that when a unitary operator is applied to either a pure or mixed state, that action transforms the system but keeps the state the same. They also evolve a system through time, but the key part is that these operations are reversible, so all the information that you begin with can be retrieved when reversing the operation[3].

3 Quantum Field Theory in Curved Spacetime

3.1 From Particles to Fields

Quantum field theory (QFT) can be quite the daunting subject, but we can simplify it a little to make it digestible. QFT is a theoretical framework that reconciles QM and special relativity, pioneered by Paul Dirac. It is a quantum description of the location of matter, and is generalized to cover all subatomic forces like the strong/weak nuclear force and electromagnetism[6].

For a conceptual understanding, we can imagine a field everywhere in the universe, then the particles that we see can be represented by localized vibrations in the field. So according to QFT, the subatomic world is just a multitude of little fields $\phi(x)$, representing up and down quark fields, and electron fields, where they are all small localized vibrations. Particle interaction would work similarly, where if we start with an electron that emits a photon, we can view it as the electron field losing some of it's energy and creating another localized photon field that travels away from the electron[6].

The field is not just a function, but an operator that can create or destroy particles and satisfy commutation or anticommutation relations. It is defined by a Lagrangian density $\mathcal{L}$, and various symmetries, following Lorentz invariance and gauge symmetry. The field can be calculated by analyzing the Fourier transform of the field operator in position space:

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2p^0}} \left(a(\vec{p})e^{ipx} + a^\dagger(\vec{p})e^{-ipx} \right) \tag{7}$$

Particles, or creation and annihilation operators, are the Fourier modes of the field operator in position space[6].

3.2 What Does QFT in Curved Spacetime Do

We have described QFT in flat spacetime, but what about in curved spacetime? The particle definition can be split into positive frequencies, which are particles, and negative frequencies, which are antiparticles. This split is based on how the time coordinate is defined. In flat spacetime, there exists a global time, giving us a unique vacuum. However, in curved spacetime, there is no universal time, so there is ambiguity in how the particles and vacuum are defined[2], [7].

Different observers mean the modes have different decompositions. The operators are related by Bogoliubov transformation, which is a mixing of creation and annihilation operators[2], [7]. Suppose two observers define the "in" mode versus "out" mode of the particles differently, their operators are related by:

$$a_{out} = \alpha a_{in} + \beta a_{in}^{\dagger} \tag{8}$$

If $\beta \neq 0$, the annihilation operator contains a creation piece, so the vacuum of one observer contains particles for another observer, leading to a mismatch in the vacuums. This means vacuum isn't invariant, and the particle number is observer-dependent, empty space is no longer a universal concept. This ties into the Unruh effect, where an inertial observer may see a vacuum, but an accelerated observer would see a thermal bath that follows a thermal distribution[2], [7]. The temperature would be tied to the acceleration a, of the observer,

$$T = \frac{\hbar a}{2\pi c k_B} \tag{9}$$

leading to the accelerated observer to have a horizon where they cannot access signals from beyond that, known as the Rindler horizon.

In semiclassical gravity, the metric describing the background is chosen first, in this case being the classical metric from GR, while the matter fields exists on this curved background. This is what Hawking's calculation approximated to describe the region surrounding a black hole. This assumes that spacetime is smooth and valid when the curvature is smaller than the Planck scale, but this may fail near singularities or during late-stage evaporation.

4 Hawking Radiation and Information Loss

4.1 Hawking Radiation Mechanism

Consider a collapsing star forming a black hole, we have "in" vacuum at past infinity and "out" particles at future infinity, the spacetime curvature distorts the field modes. The mode evolution mixes positive and negative frequencies, the particle and antiparticle modes, as written in the Bogoliubov transformation in equation 8, with $\beta \neq 0$ leading to particle

creation[2], [7].

When these particles are created, the outgoing mode is entangled with the interior mode inside the event horizon in a pure entangled state[2], [8]:

$$|\Psi\rangle \sim \sum_n e^{-n\omega/2T} |n\rangle_{out} |n\rangle_{in} \tag{10}$$

That means that near the event horizon, quantum fields are in their vacuum state but since the spacetime is curved, the vacuum isn't empty but contains these fluctuations. Pair modes are produced near the horizon, and sometimes one will fall into the black hole, while the other escapes to infinity as radiation. But like before, these modes are still entangled at creation between outside and inside, but globally with the radiation and interior the system is in a pure state[2], [8].

4.2 Thermal Nature of Radiation

The Bekenstein-Hawking entropy describes the thermal nature of Hawking radiation:

$$S_{BH} = \frac{k_B c^2 A}{4G\hbar} \tag{11}$$

An outside observer would not be able to access this information inside the interior, so we must trace over the interior degrees of freedom. The entropy arises from this, where the radiation appears thermal due to the loss of information about correlations with the interior. We can get a reduced density matrix that's proportional to a thermal distribution, giving a black body spectrum. From this, we can follow Hawking's derivation to get the Hawking temperature T_H:

$$T_H = \frac{\hbar c^3}{8\pi G M k_B} \tag{12}$$

where the temperature is only dependent on mass, or in general, on the charge and angular momentum as well. This goes back to the no-hair theorem where the radiation depends only on the macroscopic parameters with no dependence on the initial state nor the microstructure[4], [5]. The thermal aspect is a consequence of the entanglement and horizon structure, not classical randomness and if Hawking's process continues, then the entanglement grows until the final radiation is in a highly mixed state, providing no information[2], [8].

Each emitted quantum increases the entanglement between the radiation and the black hole since as a new pair is created and one of the particles falls in, the radiation is entangled with the interior, so the entanglement entropy increases over time. However, small corrections to Hawking's calculations won't fix this [2], but entanglement grows until the final radiation is maximally mixed.

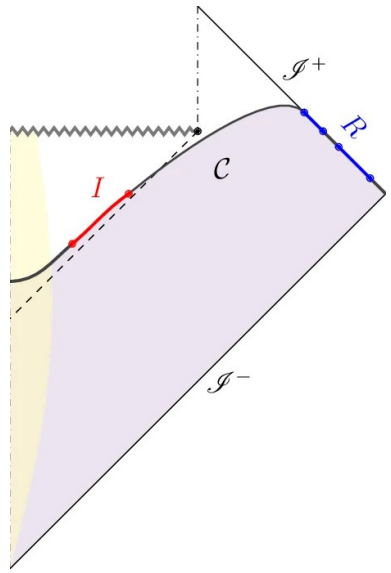

Figure 1: Penrose diagram of an evaporating black hole. Red arrow is the accretion of null dust, forming the event horizon (black dotted). When the black hole evaporates, it leaves Minkowski space. Nature of point B is unclear unless looking through a classical lens, where it is a naked singularity creating a Cauchy horizon H(red dotted). This leads to questions about the application of this diagram to unitary black hole evaporation[9].

4.3 Formulation of the Paradox

This shows that the initially pure quantum state of the infalling matter turns into a mixed state with the radiation emitted, so the exterior state is mixed. This evolution violates unitarity, which demands that the evolution of a quantum state must preserve the norm, information contained, and the state. At the end when the black hole evaporates completely, only the mixed state radiation remains. The semiclassical calculation done sparks the contradiction, violating unitarity, reversibility, and the conservation of information[3].

Additionally, if we trace the outgoing radiation backward in time and near the horizon, the wavelength shrinks exponentially and the frequency becomes extremely large, exceeding the Planck scale. This is a problem because it implies that the predicted radiation depends on unknown, short-distance physics which makes the derivation using semiclassical physics invalid with our current understanding. This is known as the trans-Planckian issue[8]. We also have another issue, the backreaction from our current assumptions. Thus far we have assumed that the Schwarzschild spacetime is fixed and that the black hole doesn't change, but in reality the Hawking radiation carries energy, making the black hole mass decrease. Backreaction is the effect of radiation on spacetime geometry, where the black hole isn't static but changes over time, which also changes the surrounding spacetime[8].

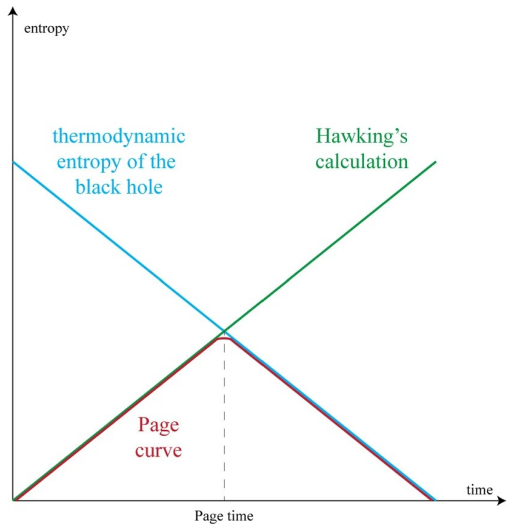

Figure 2: The Page curve of an evaporating black hole[10].

5 Entanglement and the Page Curve

5.1 Entanglement as a Measure of Information

The information contained in Hawking radiation can be understood through the concept of entanglement entropy. If the evolution of the total system, the black hole and the emitted radiation, is unitary, then the quantum state of the whole system should be pure at all times and it's von Neumann entropy remains zero. However, subsystems of a pure state can have nonzero entropy because of entanglement. Particularly, if the total state is pure then the entropy of radiation S_{rad}, and the entropy of the leftover black hole S_{BH}, should be equal[3], [11].

Depending on the entanglement entropy of the radiation, we can find out the amount of information that cannot be access in the full system by studying just the radiation. There are two cases, the first being that the radiation is highly entangled with the interior of the black hole, in which case it would appear mixed to an outside observer and carries very little accessible information. The second case is where the radiation isn't as entangled with the black hole, but correlated to other radiation, holding information about the initial state. Therefore, by studying the evolution of the radiation entropy S_{rad}, we have a quantitative way to study whether information is preserved or lost while the black hole is evaporating[3], [11].

5.2 Hawking's Prediction for Entropy Growth

In Hawking's semiclassical calculation, there is radiation produced in entangled pairs outside the event horizon, and one particle will escape to infinity as radiation while the other will fall into the black hole interior[2]. Because of this, the radiation is entangled with the interior of the black hole, with the degrees of freedom behind the horizon. By tracing over the interior modes, the outgoing radiation is described by a mixed and approximately thermal density

matrix[2], [8].

This process repeats at each step of evaporation, and so the entanglement entropy of the radiation increases over time with each pair that gets separated, adding more entanglement between the radiation and black hole. However, there is no mechanism in the semiclassical framework to reduce this correlation. Eventually the black hole would evaporate completely and the final state is just radiation that remains mixed with the degrees of freedom that don't exist anymore. So the transition from the initial pure state to the final mixed state is a direct violation of unitarity[3], [11].

5.3 The Page Curve and Unitary Evolution

In this viewpoint, the black hole evaporation is considered to be fundamentally unitary. Don Page had considered what the expected behaviour of the entanglement entropy of the radiation would be under the assumption that the whole system evolves like the usual pure quantum state. He showed that the radiation entropy should increase initially, and eventually reach a maximum at some characteristic time, called the Page time, and then decrease back to zero during the evaporation of the black hole until it's fully gone. The Page time is the point at which the black hole has radiated away exactly half of its initial coarse-grained entropy[10], [12].

During early times, the radiation can be considered a small subsystem compared to the black hole, and so is highly entangled with the interior. This gives an increase in the radiation entropy which agrees with Hawking's result. But after the Page time, the radiation becomes the larger subsystem, and according to general properties of entanglement in large quantum systems, the entropy has to decrease. This means that the new radiation emitted has to carry correlations with the earlier radiation before the Page time, which allows for the information about the initial state to be recovered eventually. Once the black hole evaporates completely, the radiation would return to a pure state and the initial information can be recovered fully[12].

5.4 Conflict Between Hawking Radiation and the Page Curve

The two predictions from the Page curve and Hawking's semiclassical result are incompatible. In Hawking's calculation, the individual emitted quantum stays entangled with it's partner mode in the interior. But according to the Page curve, unitarity requires that after the Page time the newly emitted quanta becomes entangled with the earlier radiation instead of the interior of the black hole, which decreases the entropy[2], [12].

Both requirements conflict with each other since a quantum system cannot be maximally entangled with two independent systems at once. This property is known as the monogamy of entanglement[11], [13]. This is a direct contradiction between the predictions from semiclassical QFT and the requirements from unitary quantum evolution.

The Page curve description of the black hole information paradox brings forward that this conflict must relate to one of the underlying assumptions breaking down. Either the validity of semiclassical QFT near the horizon, the unitarity of QM, or the no-hair assumption that the event horizon is smooth and featureless has to be called into question. The resolution of this paradox remains one of the main points of contention when studying quantum gravity.

6 Proposed Solutions to the Information Paradox

6.1 Black Hole Complementarity

Developed by Leonard Susskind and his collaborators, the principle of black hole complementarity was one of the first attempts at resolving the information paradox. The main idea is that there isn't one single global description of a black hole that's accessible to all observers, but instead states that the different observers might have contradictory descriptions of reality and yet not one observer would witness a violation of the laws of physics[1], [14].

To an observer at infinity, the information is never lost. Instead, the quantum information of the infalling matter is transferred to the horizon degrees of freedom at or near the event horizon and then re-emitted gradually through Hawking radiation. This process would be unitary, so the Page curve is respected. From this perspective, the horizon would behave like a physical membrane that stores and processes information. To an observer falling in, they would experience nothing unusual while crossing the horizon, obeying the equivalence principle. Information would simply pass through the horizon and continue to the interior, without being encoded or re-emitted[1], [14].

As one may conclude, black hole complementarity would say that both of these descriptions can't be combined and attempting to would lead to contradictions. But these contradictions are unphysical since neither observer would be able to see both viewpoints simultaneously. The problem with this principle stems from the monogamy of entanglement[11], [13]. As was previously stated, unitarity and the semiclassical scheme require the new radiation to be entangled with the previous radiation and the interior modes simultaneously, which cannot occur due to this principle. This is a problem that complementarity cannot resolve alone without modifying one of its underlying assumptions.

6.2 Holographic Principle and AdS-CFT Correspondence

Another approach to the paradox stems from the holographic principle, stating that information contained in a volume of spacetime can be encoded onto its boundary. This was made concrete in the AdS-CFT correspondence, which was introduced by Juan Maldacena, providing both a gravitational theory in the bulk spacetime and a non-gravitational QFT on the boundary[15].

In this case, a black hole in the bulk spacetime is equivalent to a thermal state in the boundary theory. The boundary theory is an ordinary quantum system that evolves respecting

unitarity, and so it would follow that the black hole formation and evaporation would also be unitary processes. This way, information isn't lost but instead encoded in the subtle correlations within the radiation[15].

More recently, there have been developments involving extreme surfaces and "island" prescriptions using this scheme that have successfully reproduced the Page curve[5], [11]. This shows that the entanglement entropy of the radiation increases initially, then decreases later after the Page time, consistent with both Hawking's result and unitary evolution respectively.

Even though this is quite the success, the holographic description introduces a form of nonlocality from the bulk spacetime perspective and it remains unclear how the smooth, local description of semiclassical QFT near the horizon would result in nonlocal encoding of the information as required by holography. Understanding this is an open problem in quantum gravity.

6.3 Firewall Hypothesis

The very distinctly named firewall hypothesis is a more extreme solution, challenging the widely accepted equivalence principle. It was proposed by Almheiri, Marolf, Polchinski, and Sully (AMPS), where it showcases that the three assumptions, semiclassical near the horizon, unitarity of QM, and the smoothness of the horizon from GR, cannot all exist at the same time[13].

The firewall hypothesis aims to reconcile the contradiction in the radiation before and after the Page time of preserving unitarity for the Page curve, alongside abiding semiclassical arguments beyond the horizon. To do this, the assumption of a smooth horizon is abandoned entirely, instead replacing the event horizon with a high-energy region, or a "firewall", that breaks the entanglement between the outgoing modes and the interior. In this result, a very unfortunate infalling observer would, instead of passing smoothly through the event horizon, be met with some violent, high-energy phenomena. This foregoes the foundational principle of equivalence in GR, but gains in preserving unitarity. As one would imagine, this is quite the controversial hypothesis, due to it suggesting that even in small curvatures, classical spacetime would break down[13].

6.4 Soft Hair and Quantum Hair

The soft hair theorem builds off of the no-hair theorem where black holes are characterized by only their mass, charge, and angular momentum. Instead, recent work by Hawking, Perry, and Strominger suggests that there may be additional degrees of freedom that a black hole possess, known as "soft hair". These degrees of freedom come from asymptotic symmetries of spacetime and correspond to low-energy excitation residing near or on the event horizon[16]. In contrast to the no-hair description, the quantum hair can, in principle, encode information about the infalling matter.

Instead of lost, the information would then be stored in these subtle horizon degrees of freedom and may influence the outgoing Hawking radiation. This provides a potential method for the storage of information without drastically changing the structure of spacetime. However, it is unsure if these soft hairs alone can sufficiently resolve the paradox, and it isn't yet established if this framework can reproduce the Page curve or account for the full recovery of information during evaporation[16].

7 Current Perspectives and Conclusion

Over time, the black hole information paradox has evolved from a problematic contradiction to a key guide for understanding quantum gravity. As emphasized by Marolf, instead of viewing the problem as information lost, instead it indicates that our current theoretical model is incomplete. Currently, there is general agreement that black hole evaporation is fundamentally unitary and that information is preserved, even though the conclusion continues to elude us[5]. This shift is mainly supported by the holographic principle and the AdS-CFT correspondence, where in both we have a dual quantum system which evolves unitarily[15], [16]. Additionally, there has been progress in recreating the Page curve involving extreme quantum surfaces and island prescriptions, within the semiclassical framework. This heavily supports that the entanglement entropy of Hawking radiation follows unitary evolution[11].

Even so, we still lack a complete physical picture of how the information is recovered. Instead, it hones in on the paradox, showing that somehow a resolution must reconcile competing principles. One issue is that of locality, and how encoding information on Hawking radiation may involve fundamentally nonlocal effects. It is unclear how such nonlocality comes from a supposedly local theory. Another important issue is the nature of the event horizon. The equivalence principle demands that an infalling observer should experience a smooth crossing past the horizon, yet some proposals like the firewall hypothesis, suggests that this should be revised. Determining whether the equivalence principle holds at the horizon or if new high-energy physics takes its place remains an open question.

Finally, while the Page curve describes the overall behaviour of entropy, it doesn't explain how the correlations between the early and late radiation occurs in order to carry information. Understanding what the microscopic mechanism is to encode and retrieve information from Hawking radiation is an ongoing challenge. These issues, along with the limitations of the semiclassical approximation, highlighted by the trans-Planckian problem and back-reaction, point towards the importance of a complete theory of quantum gravity. These developments indicate that the resolution of the paradox will likely require us to revise our understanding of spacetime itself. Most modern approaches, like the holographic principle and AdS-CFT correspondence, provide the mathematical background for why information is preserved and evaporation is unitary. However, AdS-CFT describes a universe that has a negative cosmological constant, or within anti-de Sitter space, which is unlike our universe that is flat. Conversely, soft hair provides the physical mechanism, answering the "how". It suggests that the vacuum is infinitely degenerate at the horizon, and so a black hole would be capable of returning the "memory" of the infalling matter with zero energy[15], [16]. By

combining these frameworks, we can guarantee that information must escape and that we have a way to carry it from holography and soft hair respectively.

In conclusion, the black hole paradox brings to light the limitations of our existing theories and while substantial progress has been made, the specific mechanism used to preserve information remains unknown. It isn't just about saving QM from GR, but is the ultimate stress-test for theoretical physics. Through the resolution of this paradox, we will gain much insight into the relationship between QM, gravity, and the structure of spacetime, making it one of the most important open problems in theoretical physics and key to a complete theory of quantum gravity.

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References

[1] L. Susskind, "The paradox of quantum black holes," *Nature Physics*, vol. 2, no. 10, pp. 665–677, 2006.

[2] S. D. Mathur, "The information paradox: A pedagogical introduction," *Classical and Quantum Gravity*, vol. 26, no. 22, p. 224 001, 2009.

[3] F. Grazioso, "Introduction to quantum information theory and outline of two applications to physics: The black hole information paradox and the renormalization group information flow," *arXiv preprint arXiv:1507.00957*, 2015.

[4] S. M. Carroll, "Lecture notes on general relativity," *arXiv preprint gr-qc/9712019*, 1997.

[5] D. Marolf, "The black hole information problem: Past, present, and future," *Reports on Progress in Physics*, vol. 80, no. 9, p. 092 001, 2017.

[6] S. Weinberg, "What is quantum field theory, and what did we think it is?" *arXiv preprint hep-th/9702027*, 1997.

[7] S. B. Giddings, "Black holes in the quantum universe," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 377, no. 2161, 2019.

[8] S. B. Giddings, "Black holes, information, and locality," *Modern Physics Letters A*, vol. 22, no. 39, pp. 2949–2954, 2007.

[9] T. Hollowood, S. Kumar, A. Legramandi, and N. Talwar, "Grey-body factors, irreversibility and multiple island saddles," Nov. 2021. DOI: `10.48550/arXiv.2111.02248`.

[10] E. Cinti and M. Sanchioni, "Beyond the quantum membrane paradigm: A philosophical analysis of the structure of black holes in full qg," *Foundations of Physics*, vol. 54, May 2024. DOI: `10.1007/s10701-024-00758-6`.

[11] S. Raju, "Lessons from the information paradox," *Physics Reports*, vol. 943, pp. 1–80, 2022.

[12] D. N. Page, "Information in black hole radiation," *Physical Review Letters*, vol. 71, no. 23, p. 3743, 1993.

[13] A. Almheiri, D. Marolf, J. Polchinski, and J. Sully, "Black holes: Complementarity or firewalls?" *Journal of High Energy Physics*, vol. 2013, no. 2, p. 62, 2013.

[14] L. Susskind, *The black hole war: My battle with Stephen Hawking to make the world safe for quantum mechanics*. Hachette+ ORM, 2008.

[15] J. Maldacena, "The large-n limit of superconformal field theories and supergravity," *International journal of theoretical physics*, vol. 38, no. 4, pp. 1113–1133, 1999.

[16] S. W. Hawking, M. J. Perry, and A. Strominger, "Soft hair on black holes," *Physical Review Letters*, vol. 116, no. 23, p. 231 301, 2016.